IE7275: Data Mining in Engineering Project 1 – Recommendation Systems for Movie Company

Project Group Members:

Tanatswa Manzini

Chan Nhi Nguyen

Yi-Fan Liu

1. Introduction

This project focuses on building and evaluating a recommendation system using real world streaming content data. The objective is to design a machine learning framework that predicts content popularity and generates structured recommendations based on historical performance patterns. Using the Netflix Movies and TV Shows dataset enriched with TMDb popularity metrics, the system analyzes relationships between genre, cast, director, budget, and engagement indicators.

Unlike traditional user focused recommender systems, this framework operates at the content strategy level. The goal is to model popularity as a measurable outcome and use predictive analytics to identify factors associated with successful content. Through systematic data preparation, exploratory analysis, model development, and evaluation, the project demonstrates how recommendation techniques can be applied to structured metadata to generate actionable insights.

2. Dataset Description

This project uses Netflix Movies and TV Shows dataset up to 2025, sourced from Kaggle and enriched with TMDb popularity metrics. The dataset contains structured metadata describing movies and television titles, including genre, cast, director, release year, budget related attributes, and engagement indicators. The dataset is suitable for recommendation modeling because it provides item level features that can be used to analyze and predict content performance.

```
print("Dataset shape:", df.shape)
print("\n")

print(df.info())
```

```
Dataset shape: (16000, 18)


<class 'pandas.core.frame.DataFrame'>
RangeIndex: 16000 entries, 0 to 15999
Data columns (total 18 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       16000 non-null  int64
 1   type          16000 non-null  object
 2   title         16000 non-null  object
 3   director      15868 non-null  object
 4   cast          15796 non-null  object
 5   country       15534 non-null  object
 6   date_added    16000 non-null  object
 7   release_year  16000 non-null  int64
 8   rating        16000 non-null  float64
 9   duration      0 non-null      float64
 10  genres        15893 non-null  object
 11  language      16000 non-null  object
 12  description   15868 non-null  object
 13  popularity    16000 non-null  float64
 14  vote_count    16000 non-null  int64
 15  vote_average  16000 non-null  float64
 16  budget        16000 non-null  int64
 17  revenue       16000 non-null  int64
dtypes: float64(4), int64(5), object(9)
```

Figure 1 Dataset Info

As shown in Figure 1, the dataset contains approximately 16,000 observations and 18 features, consisting of both categorical and numerical variables. Categorical variables include genre, cast, director, country, and language. Numerical variables include popularity, vote average, vote count, runtime, budget, and revenue. The dataset contains mixed data types and several missing values, particularly in director, cast, country, genres, and description.

The target variable in this project is popularity, treated as a continuous outcome. Popularity reflects aggregated engagement signals derived from TMDb and serves as a proxy for content performance. The modeling task is formulated as a regression problem where popularity is predicted using structured metadata features. Initial inspection involved verifying dataset dimensions, reviewing data types, checking for duplicates, and identifying missing values to ensure data integrity before preprocessing.

3. Data Preparation

The raw dataset was transformed into a structured feature matrix suitable for modeling. Preprocessing included handling missing values, removing duplicates, encoding categorical variables, transforming skewed numerical features, scaling predictors, and splitting the dataset for evaluation.

3.1 Handling Missing Values

• Step 1: Drop fully empty column duration

The column contained only missing values and was removed since it provided no usable information.

• Step 2: Remove rows missing both genres and description

Rows lacking both content descriptors were removed because they did not contain sufficient information for modeling.

• Step 3: Impute missing genres using keywords

Missing genre values were inferred using rule-based keyword matching from the description field to preserve usable observations.

• Step 4: Fill missing description

Missing descriptions were replaced with a placeholder string to maintain consistency in text processing.

• Step 5: Remove rows missing director, cast, or country

These attributes are essential for talent-based recommendation and were required for downstream modeling.

3.2 Duplicate Removal

Duplicate entries were identified using title level identifiers and removed to prevent biased frequency counts and inflated evaluation metrics.

3.3 Feature Encoding

• Directors were grouped by frequency to reduce high cardinality and sparsity.

• Cast members were split into individual actors and encoded using multi label binarization to represent multiple actors per title.

• Genres, language, and grouped directors were one hot encoded to convert categorical variables into numerical indicators suitable for modeling.

3.4 Normalization and Transformation

• Log transformation was applied to revenue and budget to reduce right skewness and stabilize variance.
• Numerical predictors such as popularity, vote count, vote average, and log budget were standardized to ensure balanced model learning across features.

3.5 Train Validation Test Split

The dataset was partitioned into 70 percent training data and 30 percent temporary data, which was further split equally into validation and testing subsets. A fixed random seed ensured reproducibility. This structure supports fair model selection and unbiased final evaluation.

3.6 Leakage Control

Outcome related variables were excluded from predictor features in supervised models to prevent target leakage and ensure realistic generalization performance.

4. Exploratory Data Analysis

Exploratory analysis was conducted to understand content engagement patterns, rating distributions, and genre level trends prior to model development. Visualizations were used to identify structural relationships and potential predictive signals.

4.1 Engagement vs Rating

A scatter plot of vote count versus vote average was generated using a logarithmic scale for vote count.

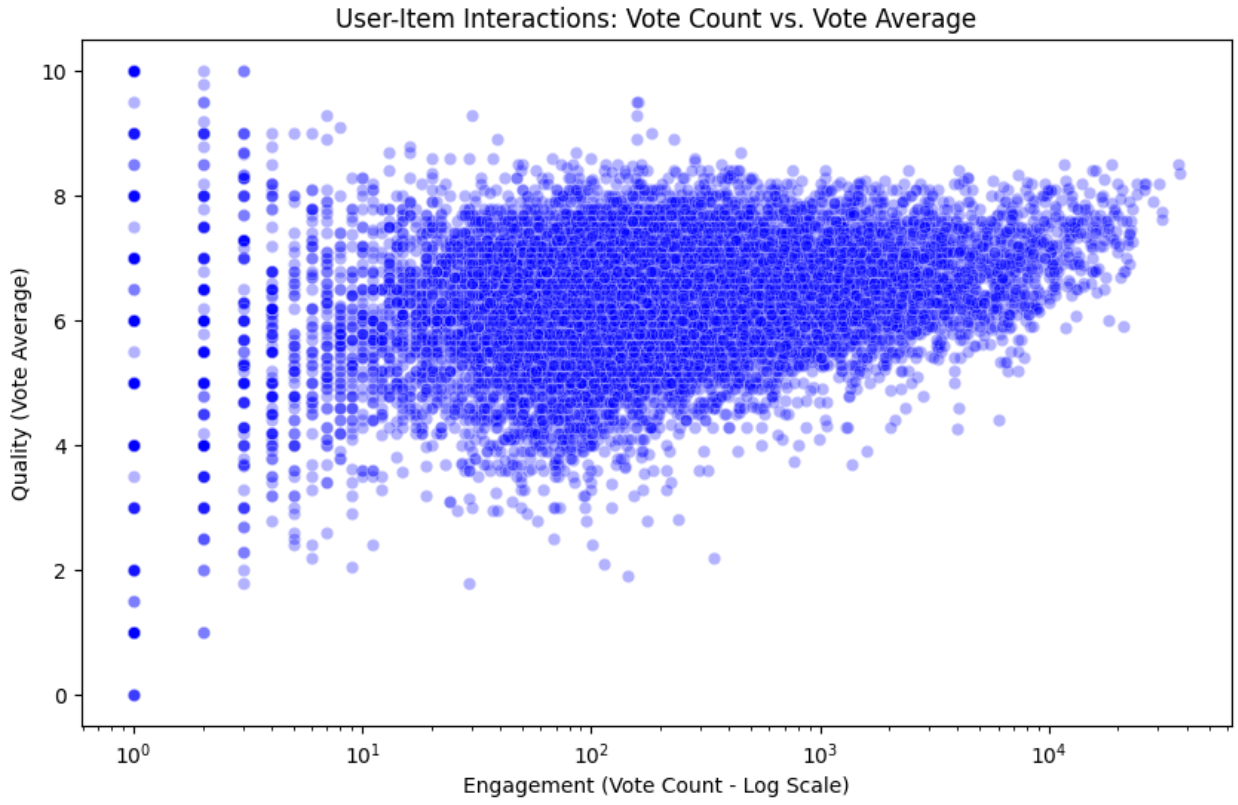


Figure 2 Vote Count vs Vote Average

This visualization shows that highly rated titles do not always correspond to high engagement levels. Several titles exhibit strong vote averages with low interaction counts, while a smaller group of titles

demonstrates both high engagement and strong ratings. This indicates that popularity is influenced by both perceived quality and exposure.

4.2 Distribution of Ratings

A histogram of vote average was produced to analyze distribution of ratings.

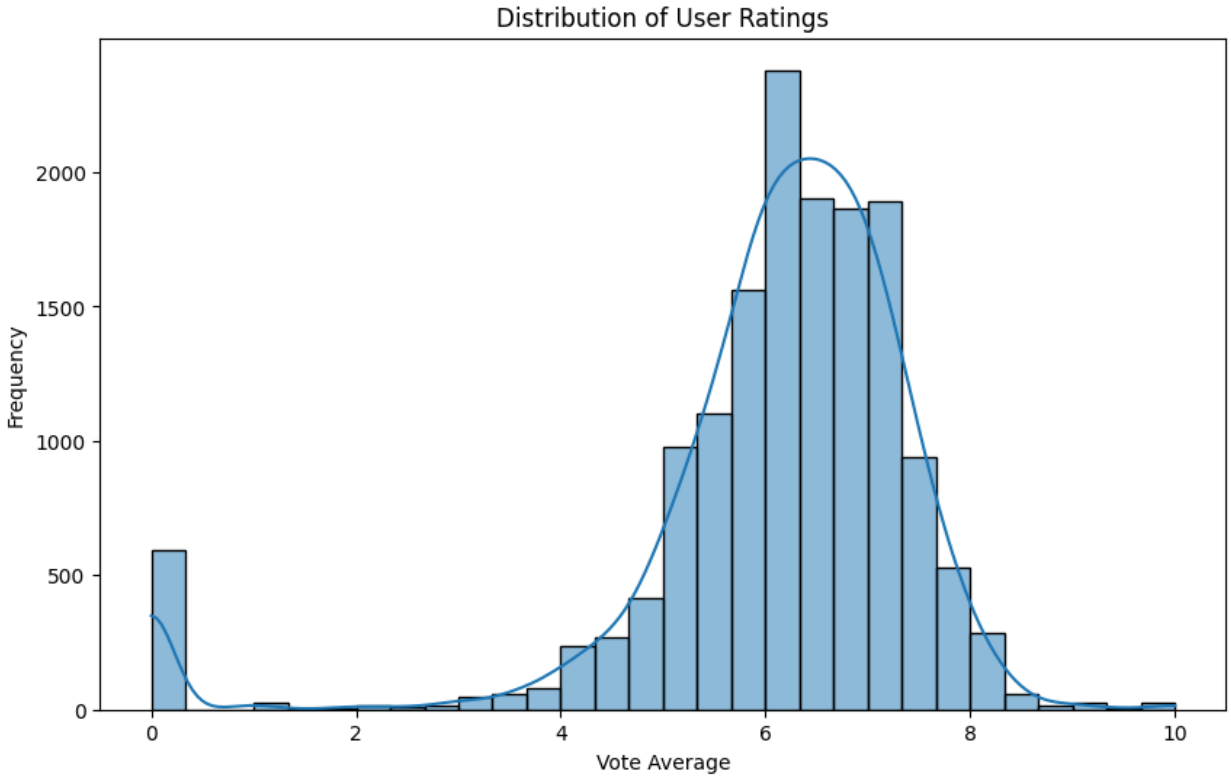


Figure 3 Distribution of Vote Average

The majority of titles cluster between ratings of 6 and 8, with relatively few extremely low or high values. This suggests limited variance in average ratings and supports the need for additional predictive signals beyond rating alone.

4.3 Genre Popularity Analysis

Average popularity was computed per genre and visualized using bar charts.

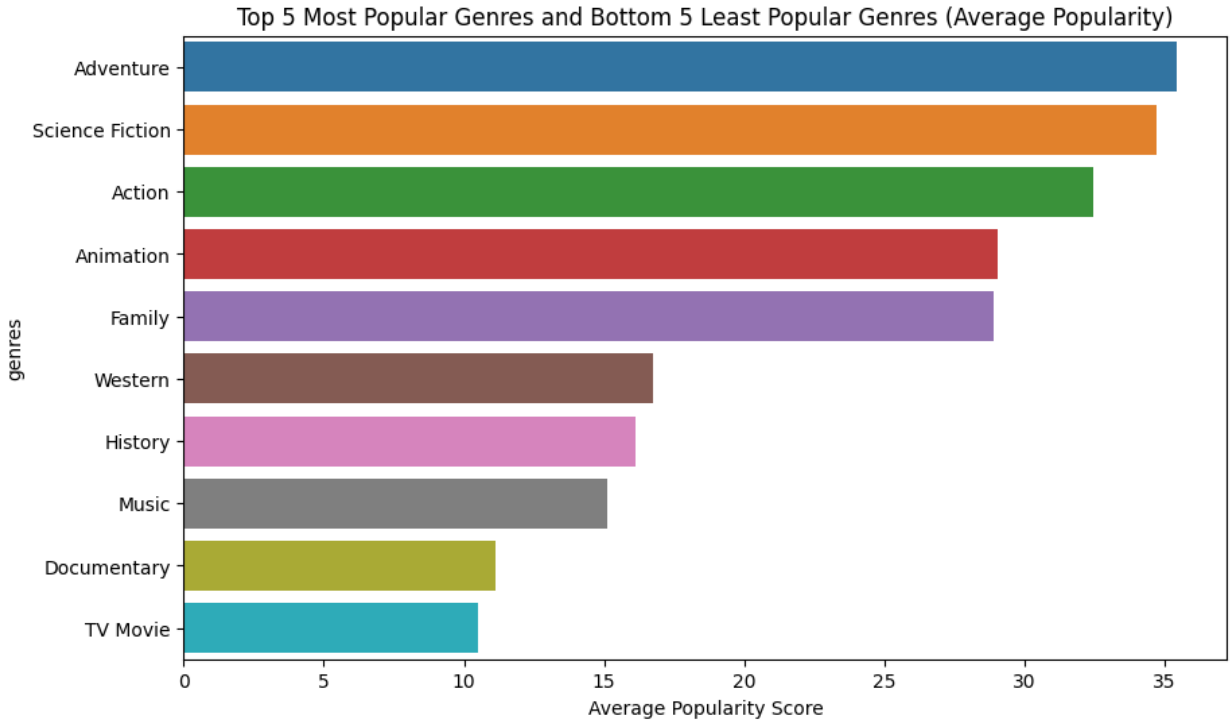


Figure 4 Average Popularity by Genre

Certain genres exhibit consistently higher average popularity levels, indicating that genre selection is a meaningful predictor variable. Lower performing genres display weaker engagement metrics.

4.4 Genre Frequency Distribution

Genre frequency counts were plotted to assess representation within the dataset.

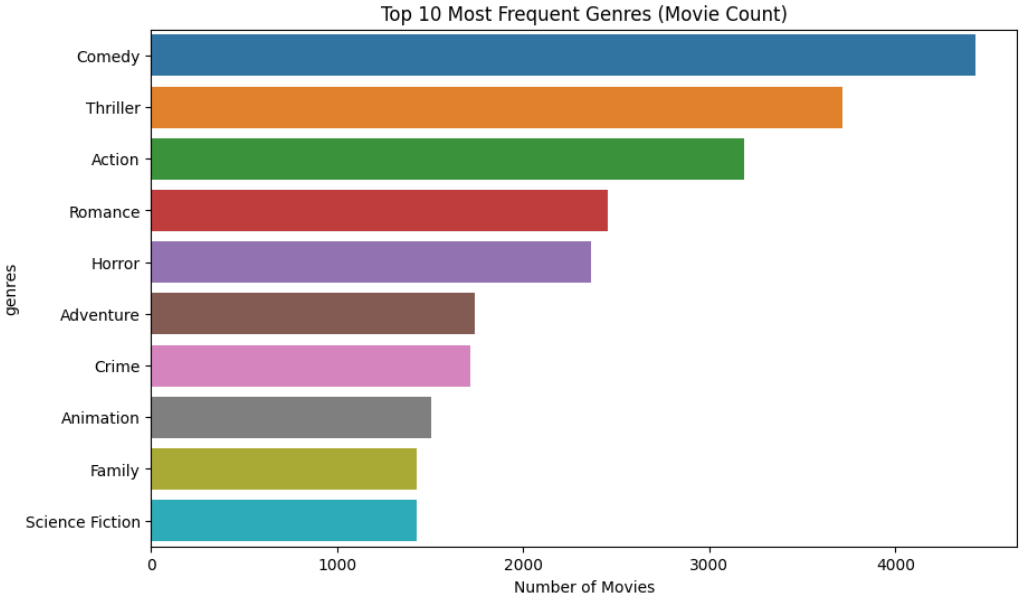


Figure 5 Genre Frequency Distribution

Some genres are significantly overrepresented compared to others. This imbalance suggests that modeling strategies must account for popularity bias and genre frequency effects.

4.5 Director and Cast Patterns

Top directors and actors were analyzed based on aggregated popularity metrics.

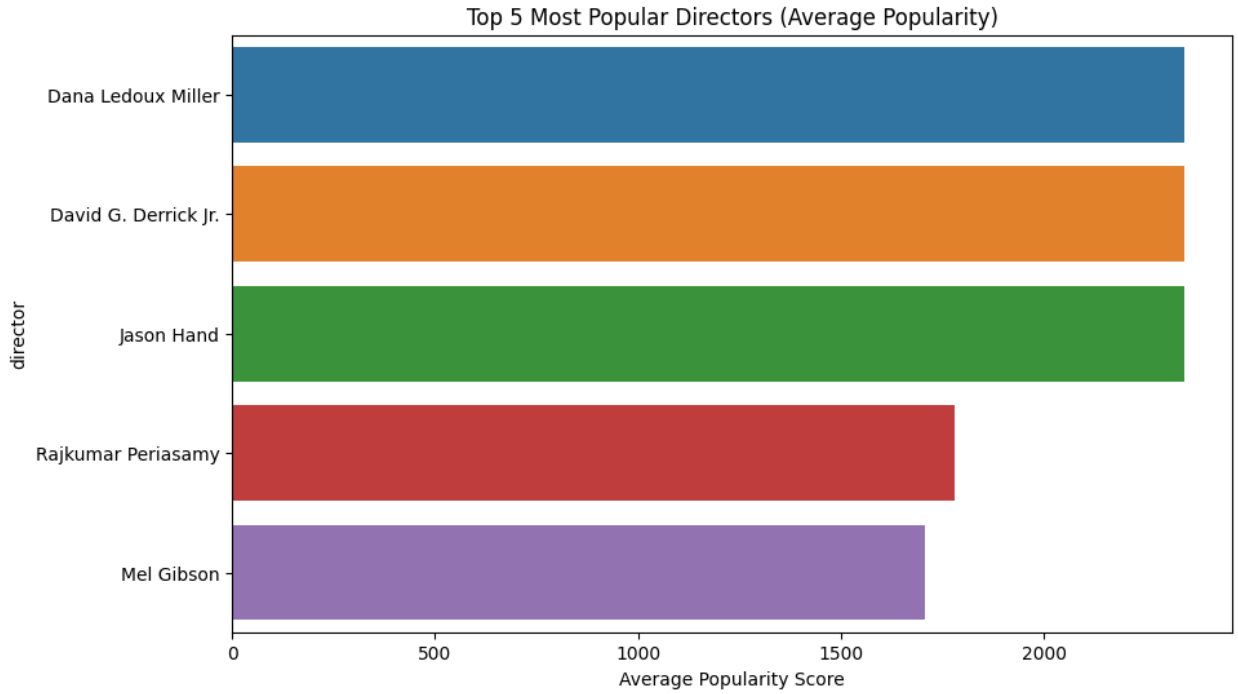


Figure 6 Top Directors by Average Popularity

A small subset of talent contributes disproportionately to high popularity outcomes. This supports the inclusion of cast and director level features in recommendation modeling.

5. Model Development

This section presents the recommendation models implemented in this project. A baseline recommendation approach was first constructed, followed by more advanced models incorporating hybrid scoring and supervised learning. Each model was evaluated using appropriate ranking and prediction metrics.

5.1 Baseline Model: Association Rule Recommendation

The baseline recommendation system was constructed using association rule mining. The Apriori algorithm was applied to transaction style representations of titles, where each transaction consisted of genres, cast members, and director information.

Frequent itemsets were identified using minimum support and confidence thresholds. Association rules were generated and used to recommend titles based on shared attributes. Recommendations were ranked using rule strength measures.

This model serves as a frequency based baseline that captures co-occurrence patterns between content attributes. It does not incorporate predictive learning and primarily reflects historical association structure.

5.2 Hybrid Model: Association + Popularity

To improve ranking quality, a hybrid scoring framework was implemented. Association rule outputs were combined with normalized popularity scores to adjust recommendation ordering.

A weighted scoring mechanism was applied to balance rule confidence and content popularity. Bayesian smoothing was used to reduce bias toward items with very few observations. This approach retains the interpretability of association rules while incorporating engagement-based refinement.

The hybrid model addresses some limitations of pure frequency-based recommendation by integrating performance signals.

5.3 Advanced Model: Supervised Success Prediction

A supervised learning model was developed using XGBoost to predict content success. Success was defined using a threshold based on log transformed revenue, representing the top performing titles.

The problem was formulated as a binary classification task. Feature inputs included encoded genres, cast indicators, director groups, language features, popularity metrics, and log transformed budget. Class imbalance was addressed using scale weighting.

XGBoost was selected due to its ability to capture nonlinear relationships, handle high dimensional structured data, and provide strong generalization performance.

Predicted probabilities were used to rank titles and generate recommendation lists. Talent level recommendations were derived by aggregating predicted success probabilities across titles associated with each actor or director.

6. Evaluation

Model performance was evaluated using ranking and classification metrics appropriate for recommendation systems. Baseline and advanced approaches were compared to assess improvements in relevance and predictive accuracy.

6.1 Ranking Evaluation

For the association and hybrid recommendation models, ranking quality was evaluated using Precision@10, Recall@10, F1@10, and NDCG@10.

Precision@10 measures the proportion of relevant titles within the top 10 recommendations. Recall@10 measures the proportion of relevant titles successfully retrieved. F1@10 balances precision and recall. NDCG@10 evaluates ranking quality by assigning higher weight to correctly ranked top positions.

The association rule model demonstrated higher recall, indicating strong coverage of frequent co-occurrence patterns. However, precision was lower due to popularity bias and broad rule generation.

The hybrid model improved ranking order by incorporating normalized popularity scores. This resulted in modest improvements in NDCG@10 and F1@10, indicating better prioritization of high-quality recommendations.

6.2 Supervised Model Evaluation

The XGBoost model was evaluated using classification and regression-based metrics.

Classification metrics included Precision, Recall, F1 score, and Area Under the Curve. The model achieved strong discriminative performance, with high AUC indicating effective separation between successful and non-successful titles.

Regression based evaluation using RMSE was also reported when modeling continuous outcomes. Lower RMSE values indicate improved prediction stability.

Predicted probabilities from the supervised model were used to generate ranked recommendation lists. Talent level rankings were derived by aggregating predicted success probabilities across associated titles.

6.3 Model Comparison

The baseline association model captures co-occurrence relationships but lacks predictive learning capability. The hybrid model improves ranking through popularity adjustment but remains limited by rule generation constraints.

The supervised model demonstrates superior predictive accuracy and ranking performance. However, it requires labeled outcome definitions and higher computational complexity.

These results highlight tradeoffs between interpretability, scalability, and predictive strength. Association based approaches offer transparency and simplicity, while gradient boosting provides stronger generalization and ranking performance.

7. Key Insights

The analysis reveals several consistent patterns across models and evaluation results.

First, genre plays a significant role in predicting popularity and success. Certain genres demonstrate consistently higher engagement metrics, indicating that content category selection materially influences performance outcomes.

Second, talent concentration effects are evident. A small subset of actors and directors is associated with disproportionately high popularity scores. This supports the inclusion of cast and director level features in predictive modeling and recommendation generation.

Third, popularity and engagement metrics provide stronger predictive signals than rating alone. While vote average exhibits limited variance, vote count and popularity measures demonstrate clearer separation between high and low performing titles.

Fourth, supervised learning substantially improves ranking precision compared to rule-based approaches. The XGBoost model captures nonlinear interactions between genres, cast composition, and budget features, leading to improved discrimination of high performing titles.

Finally, log transformation of revenue and budget stabilizes modeling performance by reducing skewness effects from extreme outliers. This confirms the importance of preprocessing decisions in predictive accuracy.

These findings demonstrate that structured metadata, when properly engineered and modeled, can provide meaningful guidance for recommendation and content evaluation tasks.

8. Limitations and Future Improvements

Several limitations should be acknowledged. First, popularity and revenue are used as proxies for content success, which may not fully capture long term audience impact or platform specific engagement dynamics. Second, the success threshold definition introduces some information compression, as continuous performance outcomes are converted into categorical labels for classification. Third, association rule models are sensitive to frequency bias, which can favor already popular genres and talent.

Additionally, the dataset does not incorporate temporal dynamics such as release seasonality or shifting audience trends over time. External factors such as marketing expenditure, platform promotion strategies, and regional preferences are also not included.

Future improvements could include incorporating time series features, implementing cross validation for more robust evaluation, and developing a hybrid candidate generation and reranking pipeline that combines rule-based filtering with supervised ranking. Further work could also explore diversification strategies to reduce popularity bias in recommendation outputs.

9. Conclusion

This project developed and evaluated multiple recommendation approaches using structured streaming content metadata. A baseline association rule model was implemented to capture co-occurrence patterns between genres, cast, and directors. A hybrid extension incorporated popularity-based weighting to refine ranking order. Finally, a supervised learning model using XGBoost was developed to predict content success and generate ranked recommendations based on predicted probabilities.

Evaluation results demonstrate that while association-based approaches provide interpretability and broad coverage, supervised learning methods achieve stronger predictive accuracy and ranking precision. Feature engineering steps such as categorical encoding, log transformation of skewed variables, and careful data partitioning contributed significantly to model performance and stability.

Overall, the findings show that structured metadata can support effective recommendation modeling when combined with appropriate preprocessing and evaluation frameworks. The project satisfies the requirements for dataset preparation, exploratory analysis, model experimentation, and performance comparison, demonstrating a complete end to end recommendation pipeline.

10. Data Source

Netflix Movies and TV shows till 2025:

https://www.kaggle.com/datasets/bhargavchirumamilla/netflix-movies-and-tv-shows-till-2025?select=netflix_movies_detailed_up_to_2025.csv